

STAT 20000: Elementary Statistics

Homework 4

Due: 11:59 pm, May 3, 2026

All page, section, and exercise numbers below refer to the course text *Statistics*, 4th edition, by D.Freedman, R. Pisani & R. Purves (FPP).

Reading: Chapters 11, 12 and 13 in FPP.

Q1 (6 points). (Chapter 13, Review Exercise 9, on p.235) A (six-face fair) die is rolled 10 times. Find the chance of:

- (a) getting 10 sixes.
- (b) not getting 10 sixes.
- (c) all the rolls showing 5 spots or less.
- (d) at least one six.

Q2 (4 points) True or false, and explain:

- (a) If a (six-face fair) die is rolled three times, the chance of getting at least one ace is $1/6 + 1/6 + 1/6 = 1/2$.
- (b) If a coin is tossed twice, the chance of getting at least one head is 100%.

Q3 (4 points) (Chapter 13, Review Exercise 6, on p.235) Four cards will be dealt off the top of a well-shuffled deck. There are two options:

- (a) To win \$1 if the first card is a club and the second is a diamond and the third is a heart and the fourth is a spade.
- (b) To win \$1 if the four cards are of four different suits.

Which option is better or are they the same? Explain.

Q4 (3 points) (Chapter 12, Review Exercise 2 on p.214) For women age 25–34 in the HANES5 sample, the relationship between height and income can be summarized as follows:

average height $\approx$ 64 inches, SD $\approx$ 2.5 inches
average income $\approx$ \$21,000, SD $\approx$ \$20,000, $r \approx 0.2$

What is the regression equation for predicting income from height? What does the equation tell you?

Q5 (2 points) (Chapter 11, Review Exercise 3 on p.198) Tuddenham and Snyder obtained the following results for 66 California boys at ages 6 and 18 (the scatter diagram is football-shaped):

average height at 6 $\approx$ 3 feet 10 inches, SD $\approx$ 1.7 inches,
average height at 18 $\approx$ 5 feet 10 inches, SD $\approx$ 2.5 inches, $r \approx 0.8$

- (a) Find the r.m.s. error for the regression prediction of height at 18 from height at 6.
- (b) Find the r.m.s. error for the regression prediction of height at 6 from height at 18.

Q6 (6 points) (Revision of Review Exercise 6, on p.235 in Chapter 13) A deck of cards is shuffled and the top two cards are placed face down on a table. True or false, and explain:

- (a) There is 1 chance in 52 for the first card to be the ace of clubs  A.
- (b) There is 1 chance in 52 for the second card to be the ace of diamonds  A. Or should the chance be $1/51$.
- (c) The chance of getting the ace of clubs and then the ace of diamonds is $1/52 \times 1/52$.
- (d) The chance that the first card is the ace of clubs or the second card is the ace of diamonds is $1/52 + 1/52$.